KU LEUVEN

Geavanceerde Computerarchitectuur 2nd session

Race conditions

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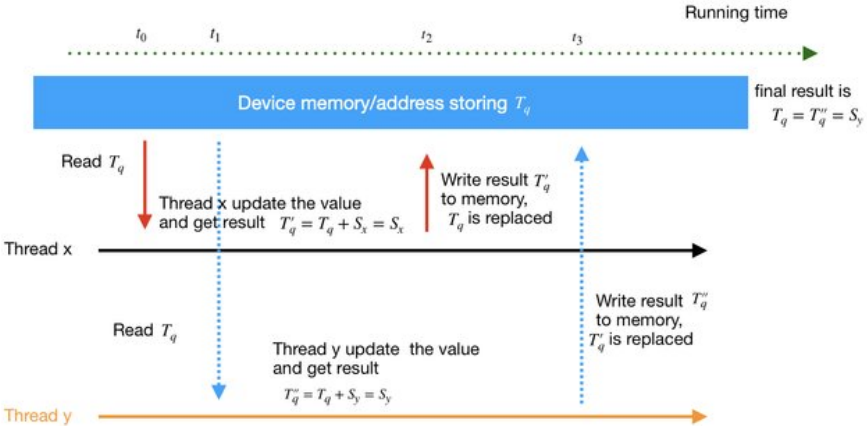
0 Outline

1. Race conditions
2. Exercise

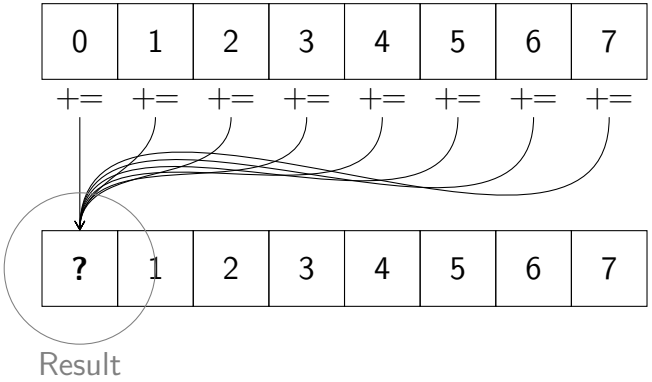
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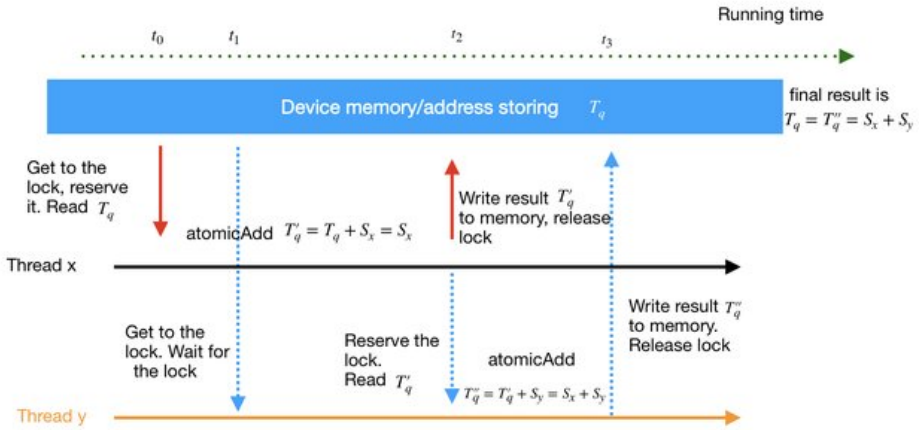
1 Race conditions



1 Addition - an example of possible race conditions



1 Resource locking and waiting as an alternative



1 Atomic operation

An atomic command gets executed without interruption

Threads get **sequential access** to a common resource

1 Atomic operation

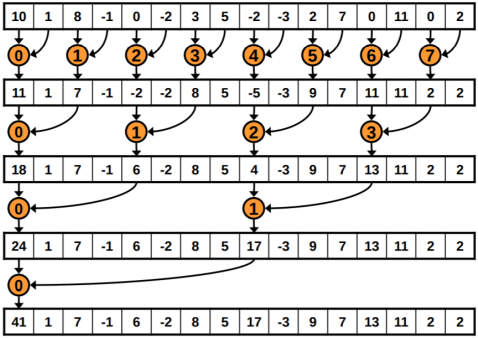
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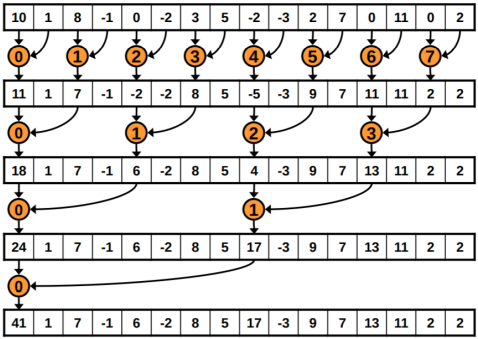
CUDA (and other interfaces) features built-in atomic functions:

- For addition `atomicAdd(...)`
- For subtraction `atomicSub(...)`
- For minimum detection `atomicMin(...)`
- For maximum detection `atomicMax(...)`
- ... full list and reference on p. 187–190 of the programming guide

1 Semi-parallel addition using reduction

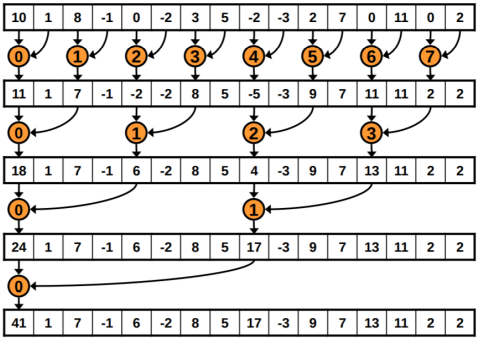


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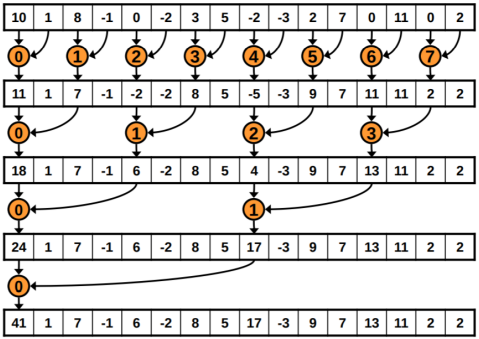
- Ideal for 2^N long inputs

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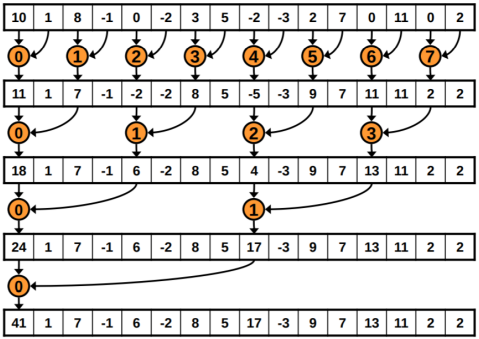
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- Shifting direction not important

2 Outline

1. Race conditions
2. Exercise

2 Max detection using three approaches

Implement max detection using:

- A nested for loop on the CPU,
- A GPU kernel employing atomic operation, and
- A GPU kernel using reduction.

Use a 1D input array of your liking.

Assess the dependency of the execution time on the input array's length.

Timing examples are provided in the cheat sheet.

2 Guiding questions

- **Is the result correct?**
- **How does time increase as a function of input size?**
- What happens if the input is larger than 2048 (how would you ensure correct calculation and are there additional synchronization steps)?
- Consider calling the kernel with much less threads than there are data elements. For example, one tenth. How would you ensure all elements get processed?
- How would you tackle a 2D array?